```
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'Project 7
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' {$STAMP BS2}
' {$PBASIC 2.5}

x           VAR Word
y           VAR Word
row         VAR Nib
column      VAR Nib
keypad      VAR Word
temp        VAR Nib
position1   VAR Word
Dball       VAR Word
start_game  VAR Word
game_mode   VAR Word
counter     VAR Word
sec         VAR Word
mill        VAR Word

mill = 0
sec = 0
counter = 0
position1 = 129
Dball = 5

SEROUT 14, 84, [22, 12, 17]
SEROUT 14, 84, [208]
SEROUT 14, 84, [248, %11111, %11111, %11111, %11111, %11111, %11111, %11111, %11111]
SEROUT 14, 84, [249, %00000, %00000, %00000, %00000, %00000, %00000, %00000, %00000]
SEROUT 14, 84, [251, %00110, %00110, %00000, %00000, %00000, %00000, %00000, %00000]
SEROUT 14, 84, [252, %00000, %00000, %00110, %00110, %00000, %00000, %00000, %00000]
SEROUT 14, 84, [253, %00000, %00000, %00000, %00000, %00110, %00110, %00000, %00000]
SEROUT 14, 84, [254, %00000, %00000, %00000, %00000, %00000, %00000, %00110, %00110]
SEROUT 14, 84, [255, %00000, %00000, %00000, %00000, %00110, %00110, %00000, %00000]


GOTO start

start:
      counter = 0
      game_mode = 0
      SEROUT 14, 84, [12, 17]
      PAUSE 10
      SEROUT 14, 84, [133, "Maze Game"]
      SEROUT 14, 84, [153, "(1) Easy", 173, "(2) Hard", 193, "(*) Start"]
      DO
      GOSUB ReadKeypad
      IF keypad THEN
      GOSUB display
      ENDIF
      LOOP
      RETURN


wrong:
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [128, "Wrong number/letter", 148, "Please try again"]
      PAUSE 100
      GOTO start
      RETURN

layout:
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [128, 1," ", 0, " ", " ", " ", 0, " ", " ", " ", 0, " ", " ", " ", 0, " ", " ","
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 ",0]
      SEROUT 14, 84, [148, 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0,
 " "]
      SEROUT 14, 84, [168, 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0,
 " "]
      SEROUT 14, 84, [188, 0, " ", " ", " ", 0, " ", " ", " ", 0, " ", " ", " ", 0, " ", " ", " ", 0,
 " ", " ", " "]
      IF counter = 0 THEN
      SEROUT 14, 84, [128, 7]
      ENDIF
      RETURN


game:
      mill = 0
      sec = 0
      DO
      PULSIN 13, 1, x
      PULSIN 12, 1, y
      x = x / 125
      y = y / 125
      PAUSE 50
      IF (counter = 0) THEN
      DO
      PULSIN 13, 1, x
      PULSIN 12, 1, y
      x = x / 125
      y = y / 125
      IF (IN13 = 0 AND IN12 = 1) THEN
      position1 = 129
      Dball = 5
      SEROUT 14, 84, [128, 1, 5]
      SEROUT 14, 84, [218, 225]
      counter = 1
      PAUSE 50
      GOTO game
      ENDIF
      LOOP
      ENDIF

      IF (position1 = 129 OR position1 = 131 OR position1 = 132 OR position1 = 133 OR position1 =
134 OR position1 = 136 OR position1 = 137 OR position1 = 138 AND Dball = 3 AND y < 19) THEN
      PAUSE 50
      GOTO lose
      ENDIF
      IF (position1 = 140 OR position1 = 141 OR position1 = 142 OR position1 = 144 OR position1 =
145 OR position1 = 146 AND Dball = 3 AND y < 19) THEN
      PAUSE 50
      GOTO lose
      ENDIF
      IF (position1 = 189 OR position1 = 190 OR position1 = 191 OR position1 = 193 OR position1 =
194 OR position1 = 195 OR position1 = 197 OR position1 = 198 AND Dball = 6 AND y > 21) THEN
      PAUSE 50
      GOTO lose
      ENDIF
      IF (position1 = 199 OR position1 = 201 OR position1 = 202 OR position1 = 203 OR position1 =
205 OR position1 = 206 OR position1 = 207 AND  Dball = 6 AND y > 21) THEN
      PAUSE 50
      GOTO lose
      ENDIF

      IF (IN13 = 0 AND IN12 = 1) THEN
      SEROUT 14, 84, [position1, " "]
      SEROUT 14, 84, [218, 225]
      position1 = position1 + 1
      ELSEIF (IN13 = 1 AND IN12 = 0) THEN
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      IF (game_mode = 2) THEN
      GOTO lose
      ENDIF
      SEROUT 14, 84, [position1, " "]
      SEROUT 14, 84, [218, 222]
      position1 = position1 - 1
      ENDIF

      IF (IN13 = 0 AND IN12 = 0) THEN
      IF (game_mode = 2) THEN
      IF (position1 = 131 OR position1 = 151 OR position1 = 171 OR position1 = 135 OR position1 =
155 OR position1 = 175 OR position1 = 139 OR position1 = 159 OR position1 = 179) THEN
      GOTO lose
      ELSEIF (position1 = 143 OR position1 = 163 OR position1 = 183 OR position1 = 167 OR position1 =
 187) THEN
      GOTO lose
      ENDIF
      ENDIF
      IF Dball = 3 THEN
      Dball = 4
      ELSEIF Dball = 4 THEN
      Dball = 5
      ELSEIF Dball = 5 THEN
      Dball = 6
      ELSEIF Dball = 6 THEN
      Dball = 3
      SEROUT 14, 84, [position1, " "]
      position1 = position1 + 20
      ENDIF
      SEROUT 14, 84, [218, 226]
      ENDIF

      IF (IN13 = 1 AND IN12 = 1) THEN
      IF (game_mode = 2) THEN
      IF(position1 = 149 OR position1 = 169 OR position1 = 189 OR position1 = 153 OR position1 = 173
OR position1 = 193 OR position1 = 157 OR position1 = 177 OR position1 = 197) THEN
      GOTO lose
      ELSEIF (position1 = 161 OR position1 = 181 OR position1 = 201 OR position1 = 165 OR position1 =
 185 OR position1 = 205) THEN
      GOTO lose
      ENDIF
      ENDIF
      IF Dball = 3 THEN
      Dball = 6
      SEROUT 14, 84, [position1, " "]
      position1 = position1 - 20
      ELSEIF Dball = 4 THEN
      Dball = 3
      ELSEIF Dball = 5 THEN
      Dball = 4
      ELSEIF Dball = 6 THEN
      Dball = 5
      ENDIF
      SEROUT 14, 84, [218, 229]
      ENDIF

      mill = mill + 1
      IF (mill > 9) THEN
      sec = sec + 1
      mill = mill - 9
      ENDIF

      IF (position1 = 130 OR position1 = 134 OR position1 = 138 OR position1 = 142 OR position1 =
146 OR position1 = 148 OR position1 = 150 OR position1 = 152 OR position1 = 154 OR position1 = 156)
THEN
      GOTO lose
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      ELSEIF (position1 = 158 OR position1 = 160 OR position1 = 162 OR position1 = 164 OR position1 =
 166 OR position1 = 168 OR position1 = 170 OR position1 = 172 OR position1 = 174) THEN
      GOTO lose
      ELSEIF (position1 = 176 OR position1 = 178 OR position1 = 180 OR position1 = 182 OR position1 =
 184 OR position1 = 186 OR position1 = 188) THEN
      GOTO lose
      ELSEIF (position1 = 192 OR position1 = 196 OR position1 = 200 OR position1 = 204) THEN
      GOTO lose
      ENDIF

      IF (position1 = 147) THEN
      GOTO win
      ENDIF
      SEROUT 14, 84, [8]
      SEROUT 14, 84, [position1, Dball]

      LOOP
      RETURN

lose:
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [150, "Loser Try Again!!"]
      PAUSE 200
      SEROUT 14, 84, [218, 227]
      PAUSE 100
      SEROUT 14, 84, [218, 226]
      PAUSE 100
      SEROUT 14, 84, [218, 225]
      PAUSE 100
      GOTO start
      RETURN

win:
      mill = mill * 5
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [150, "You Won Smarty!!"]
      SEROUT 14, 84, [171, "Your Time is", 192, DEC3 sec * 42/13 , ".", 197, DEC2 mill, 200, "sec"]
      PAUSE 2000
      SEROUT 14, 84, [218, 220]
      PAUSE 100
      SEROUT 14, 84, [218, 223]
      PAUSE 100
      SEROUT 14, 84, [218, 228]
      PAUSE 100
      GOTO start
      PAUSE 2000
      RETURN

ReadKeypad:
      keypad = 0
      OUTL = %00000000
      DIRL = %00000000
      FOR row = 0 TO 3
      DIRB = %1111
      OUTB = %0000
      OUTA = 1 << row
      DIRA = 1 << row
      temp = 0
      FOR column = 0 TO 3
      INPUT (column + 4)
      temp = temp | (INB & (1 << column))
      NEXT
      keypad = keypad << 4 | (Temp REV 4)
      NEXT
      RETURN
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Display:
      IF KeyPad.BIT15 THEN game_mode = 1
      IF Keypad.BIT14 THEN game_mode = 2
      IF KeyPad.BIT13 THEN GOTO wrong
      IF Keypad.BIT12 THEN GOTO wrong
      IF KeyPad.BIT11 THEN GOTO wrong
      IF Keypad.BIT10 THEN GOTO wrong
      IF KeyPad.BIT9 THEN GOTO wrong
      IF Keypad.BIT8 THEN GOTO wrong
      IF KeyPad.BIT7 THEN GOTO wrong
      IF Keypad.BIT6 THEN GOTO wrong
      IF KeyPad.BIT5 THEN GOTO wrong
      IF Keypad.BIT4 THEN GOTO wrong
      IF KeyPad.BIT3 THEN
      IF (game_mode = 1 OR game_mode = 2) THEN
      SEROUT 14, 84, [209, 218, 220]
      SEROUT 14, 84, [209, 222]
      SEROUT 14, 84, [209, 223]
      SEROUT 14, 84, [209, 225]
      SEROUT 14, 84, [209, 227]
      GOSUB layout
      GOTO game
      ELSE
      GOTO wrong
      ENDIF
      IF Keypad.BIT2 THEN GOTO wrong
      IF KeyPad.BIT1 THEN GOTO wrong
      IF Keypad.BIT0 THEN GOTO wrong
      ENDIF
      RETURN
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